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Department of Mathematics
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Stat 201 Major Exam I Term 251

Original exam date: 29 September 2025. Practice version: questions and choices shuffled.
Net time allowed: 90 minutes

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1. (Exercise 2-95) A lot of 40 semiconductor chips contains 5 that are defective. Two are selected randomly, without replacement, from the lot. What is the probability that the second one selected is defective given that the first one was defective?

(a) $\frac{1}{78}$

(b) $\frac{4}{39}$

(c) $\frac{5}{39}$

(d) $\frac{1}{8}$

(e) $\frac{1}{10}$

2. (Exercise 3-42) Errors in an experimental transmission channel are found when the transmission is checked by a certifier that detects missing pulses. The number of errors found in an eight-bit byte is a random variable with the following distribution:

$$F(x) = \begin{cases} 0, & x < 0, \\ 0.45, & 0 \le x < 3, \\ 0.62, & 3 \le x < 5, \\ 0.90, & 5 \le x < 8, \\ 1, & 8 \le x. \end{cases}$$

What is the probability that there are 4 errors in an eight-bit byte, i.e., $P(X = 4)$?

(a) 0

(b) 0.28

(c) 0.17

(d) 0.45

(e) 0.62

3. (Exercise 3-175) From 500 customers, a major appliance manufacturer will randomly select a sample without replacement. The company estimates that 30% of the customers will provide useful data. If this estimate is correct, what is the probability mass function of the number of customers that will provide useful data? Assume that the company samples 5 customers.

(a) Poisson with $\lambda = 1.5$

(b) Hypergeometric with $n = 150$, $K = 5$ and $N = 500$

(c) Hypergeometric with $n = 5$, $K = 150$ and $N = 500$

(d) Binomial with $n = 500$ and $p = 0.3$

(e) Binomial with $n = 5$ and $p = 0.3$

4. (Exercise 3-29) An assembly consists of three mechanical components. Suppose that the probabilities that the first, second, and third components meet specifications are 0.92, 0.88, and 0.83, respectively. Assume that the components are independent. Determine the probability mass function of X: the number of components in the assembly that meet specifications.

(a)

x	1	2	3
$P(X = x)$	0.038016	0.278784	0.681472

(b)

x	1	2	3
$P(X = x)$	0.034656	0.291744	0.671968

(c)

x	0	1	2	3
$P(X = x)$	0.004913	0.071961	0.351339	0.571787

(d)

x	0	1	2	3
$P(X = x)$	0.001632	0.038704	0.287696	0.671968

(e)

x	0	1	2	3
$P(X = x)$	0.001632	0.287696	0.038704	0.671968

5. (Exercise 6-55) The twelve measurements that follow are furnace temperature recorded on successive batches in a semiconductor manufacturing process (units are °C): 498.17, 504.23, 504.83, 510.30, 511.35, 511.36, 511.46, 514.20, 514.84, 518.37, 529.91, 535.49. Using the Interquartile Range (IQR) method, which one of the following statements about outliers in this dataset is correct?

(a) 498.17 is the only outlier.

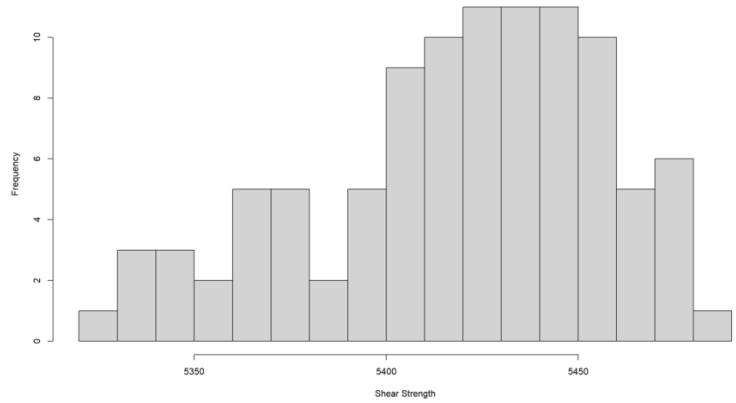
(b) 498.17 and 535.49 are the two outliers.

(c) There is no outlier.

(d) 529.91 and 535.49 are the two outliers.

(e) 535.49 is the only outlier.

6. (Exercise 6-47) The shear strengths of 100 spot welds in a titanium alloy are recorded. A histogram of these data is shown in the figure below. Which one of the following statements about this histogram is true?



(a) The mean shear strength is around 5350.

(b) The median shear strength is around 5350.

(c) The data is negatively skewed.

(d) The data is symmetric.

(e) The data is positively skewed.

7. (Suggested Problem 2-147) Customers are used to evaluate preliminary product designs. In the past, 90% of highly successful products received good reviews, 70% of moderately successful products received good reviews, and 20% of poor products received good reviews. In addition, 50% of products have been highly successful, 35% have been moderately successful, and 15% have been poor products. If a product does not attain a good review, what is the probability that it will be a highly successful product?

(a) 0.1001

(b) 0.5922

(c) 0.6207

(d) 0.3818

(e) 0.1818

8. (Exercise 6-6) Suppose that you add 7 to all of the observations in a sample. How does this change the sample mean? How does it change the sample standard deviation?

(a) Both the sample mean and the sample standard deviation are unchanged.

(b) Both the sample mean and the sample standard deviation increase by 7.

(c) The sample mean increases by 7; the sample standard deviation is unchanged.

(d) The sample mean increases by 7; the sample standard deviation decreases by 7.

(e) The sample mean is unchanged; the sample standard deviation increases by 7.

9. (Exercise 2-79) The analysis of shafts for a compressor is summarized by conformance to specifications.

	Roundness conforms	
Surface finish	Yes	No
Yes	329	18
No	17	12

What is the probability that the selected shaft either conforms to surface finish requirements or does not conform to roundness requirements?

(a) 0.9069

(b) 0.1277

(c) 0.9234

(d) 0.0798

(e) 0.9548

10. (Exercise 3-89) A particularly long traffic light on your morning commute is green 10% of the time that you approach it. Assume that each morning represents an independent trial. Over 30 mornings, what is the probability that the light is green on more than two days?

(a) 0.5886

(b) 0.2361

(c) 0

(d) 0.2277

(e) 0.4114

11. (Exercise 3-138) The number of flaws in bolts of cloth in textile manufacturing is assumed to be Poisson distributed with a mean of 0.22 flaw per square meter. What is the probability that there is one flaw in 9 square meters of cloth?

(a) 1.589

(b) 0.273

(c) 0.862

(d) 0.177

(e) 0.244

12. (Exercise 2-56) Orders for a computer are summarized by the optional features that are requested as follows: no optional features, proportion 0.18; one optional feature, proportion 0.62; more than one optional feature, proportion 0.20. What is the probability that an order requests at least one optional feature?

(a) 0.820

(b) 0.380

(c) 0.200

(d) 0.124

(e) 0.620

13. (Suggested Problem 2-29) The rise time of a reactor is measured in minutes (and fractions of minutes). Let the sample space be positive, real numbers. Define the events A and B as follows:

$$A = \{x \mid x < 45.5\}, \qquad B = \{x \mid x > 20.5\}.$$

Which of the following best describes $A \cap B$?

(a) $\{x \mid x > 0\}$

(b) $\{x \mid x \geq 45.5\}$

(c) $\{x \mid 25.5 < x < \infty\}$

(d) $\{x \mid x \leq 20.5\}$

(e) $\{x \mid 20.5 < x < 45.5\}$

14. (Exercise 2-135) Disks of polycarbonate plastic from a supplier are analyzed for scratch and shock resistance. The results from 95 disks are summarized as follows:

	Shock resistance	
Scratch resistance	High	Low
High	73	7
Low	10	5

Let A denote the event that a disk has high shock resistance, and let B denote the event that a disk has high scratch resistance. Which one of the following statements is NOT true?

(a) A and B are not mutually exclusive

(b) $P(B \mid A) = \dfrac{73}{83}$

(c) $P(A \mid B) = \dfrac{73}{80}$

(d) A and B are independent

(e) $P(A' \mid B') = \dfrac{1}{3}$

Answer Key

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Q	Ans	Brief explanation
1	**(b)**	After one defective is removed: $\frac{4}{39}$.
2	**(a)**	F is flat at $x = 4$ (no jump), so $P(X = 4) = 0$.
3	**(c)**	Without replacement: hypergeometric, $N = 500$, $K = 150$, $n = 5$.
4	**(d)**	$P(X = 0) = 0.001632$, $P(1) = 0.038704$, $P(2) = 0.287696$, $P(3) = 0.671968$.
5	**(e)**	$Q_1 = 507.565$, $Q_3 = 516.605$; upper fence 530.165, so only 535.49 is an outlier.
6	**(c)**	Long tail to the left, bulk near 5420 to 5450: negatively skewed.
7	**(e)**	$\frac{0.5(0.1)}{0.5(0.1)+0.35(0.3)+0.15(0.8)} = \frac{0.05}{0.275} \approx 0.1818$.
8	**(c)**	Mean shifts by $+7$; deviations from the mean unchanged, so s unchanged.
9	**(e)**	$P(A \cup B) = \frac{347+30-18}{376} = \frac{359}{376} \approx 0.9548$.
10	**(a)**	Binomial(30, 0.1): $1 - P(X \leq 2) = 1 - 0.4114 = 0.5886$.
11	**(b)**	$\lambda = 0.22 \cdot 9 = 1.98$; $P(X = 1) = 1.98e^{-1.98} \approx 0.273$.
12	**(a)**	$1 - 0.18 = 0.62 + 0.20 = 0.820$.
13	**(e)**	Overlap of $x < 45.5$ and $x > 20.5$ is $20.5 < x < 45.5$.
14	**(d)**	$P(A)P(B) \approx 0.736 \neq 0.768 = P(A \cap B)$, so independence is false.